

High-Mobility Wireless Channel Measurements at 5.9 GHz in an Urban Environment

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Abstract—Next-generation intelligent transportation systems require a communication technology that can satisfy high demands in terms of latency and reliability. One of the promising communication technologies to satisfy such demands is the fifth-generation (5G) new radio (NR) for the vehicle-to-everything (V2X). To support the development of NR-V2X systems in the 5.9 GHz band, it is essential to perform channel measurements in high-mobility scenarios. High-mobility scenarios in the 5.9 GHz band have been well investigated through drive-by measurement campaigns. Although such measurements deliver real-world results, they do not allow for controlled experiments. To investigate propagation characteristics of the 5.9 GHz band, we perform controllable and repeatable channel measurements at different velocities (25, 50, 100 and 125 km/h) in an urban environment. Furthermore, we compare measured wireless channels in terms of the normalized average local scattering function.

Index Terms—channel sounding, measurements, testbed, 5G, NR-V2X, vehicular communications.

I. INTRODUCTION

Future intelligent transportation systems (ITS) will offer a wide range of vehicular applications to increase road safety [1] and improve the traffic flow in urban environments [2]. To achieve the objectives mentioned above, the vehicle-to-everything (V2X) communication technology should ensure very high reliability and ultra-low latency [3]. A promising communication technology to support various vehicular applications is the fifth generation (5G) new radio (NR) for the V2X. One of the deployment bands for NR-V2X communications is the ITS unlicensed band at 5.9 GHz [4]. To support the design and development of NR-V2X systems in the 5.9 GHz band, it is crucial to perform channel measurements in high-mobility scenarios. So far, the 5.9 GHz band has been well investigated through drive-by vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) measurement campaigns [5]–[20]. Although such measurements deliver real-world results, they do not allow for controlled experiments. Since a typical vehicular environment consists of moving vehicles and pedestrians (scattering objects), one cannot control environmental and system parameters. Furthermore, drive-by measurement campaigns do not allow repeating a measurement sample under identical channel conditions. As a consequence of the altered

measurement environment, one can obtain different large and small-scale fading effects.

The rest of the paper is organized as follows. Section II describes our measurement setup for the 5.9 GHz band. The measurement campaign and methodology are described in Section III, whereas Section IV presents the obtained results. Finally, Section V concludes the paper.

Contribution: We adapt the indoor-to-indoor testbed hardware proposed in [21] enabling indoor-to-outdoor measurements in a repeatable and controlled high-mobility environment. Furthermore, we conduct wireless channel measurements at 5.9 GHz in an urban environment using the adapted testbed. We employ a moving transmit antenna and a static receive antenna. Measurements are performed at different velocities (25, 50, 100 and 125 km/h) and for an identical trace of the transmit antenna. Finally, we compare and discuss measurement results in terms of an estimated local scattering function of the channel.

II. TESTBED ARCHITECTURE

Our measurement testbed (see Fig. 1) consists of a moving transmitter and a static receiver. The moving transmitter is based on a rotary unit described in [22]. The rotary unit rotates the transmit antenna around a central axis with a constant but adjustable speed. The transmit antenna is connected to a rotary joint by a coaxial cable through a 1 m long rotary arm. The rotary joint is used to feed a transmit signal from a signal source to the rotating transmit antenna.

We use an orthogonal frequency division multiplexing (OFDM) signal as a transmit signal. To keep the peak-to-average power ratio (PAPR) of the signal low, the transmit signal is designed using Zadoff-Chu sequence [23]. A transmit symbol is given by

$$x[n] = \text{Re} \left(\sum_{k=-K/2}^{K/2-1} e^{-j\pi \frac{k^2}{K}} e^{j2\pi \frac{kn}{N}} \right), \quad (1)$$

where n denotes the time-index, k is the subcarrier index, K represents the number of subcarriers and N is the number of samples per OFDM symbol.

The transmit OFDM signal is generated off-line using Python and digitally up-converted to $f_{RF} = 5.9$ GHz by an Arbitrary Waveform Generator (Keysight M8195A). After that, the up-converted signal is fed through the rotary joint

This work was supported by the Austrian Research Promotion Agency (FFG) via the research project Intelligent Intersection (ICT of the Future, Grant 880830) and the Christian Doppler Laboratory for Dependable Wireless Connectivity for the Society in Motion.

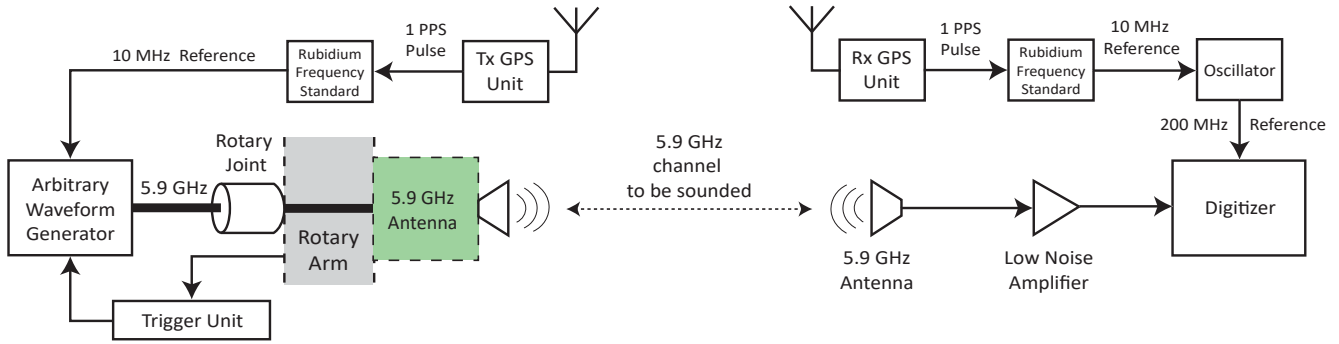


Fig. 1. Channel sounding hardware setup for the 5.9GHz indoor-to-outdoor scenario.

and emitted by a self-built 5.9GHz omnidirectional monopole antenna (see Fig. 2).

At the receiver side, we employ a vertically polarized omnidirectional antenna (Smarteq VO9/54) (see Fig. 3) and a low-noise amplifier (Mini Circuits ZJL-6G+). By selecting omnidirectional antennas at both transmit and receive sides, we enhance the multi-path characteristics of a wireless channel. Finally, we sample the received signal with a signal analyzer (Guzik Digitizer ADP7104), which acts as a receiver.

III. MEASUREMENT CAMPAIGN

We perform indoor-to-outdoor high-mobility wireless channel measurements in the inner city of Vienna, Austria. The measurement environment is shown in Fig. 4. The transmitter is placed on the fifth floor inside the building and moves on a circular arc with a constant velocity at the rotary unit. The receiver is placed 140m apart from the transmitter and is static on the roof of another building in the center of Vienna.

Since the receiver is placed outdoors far away from the transmitter, it is not easily feasible to interconnect the arbitrary waveform generator at the transmitter and the signal analyzer at the receiver by a cable to provide a precise frequency and time synchronization as in [21]. As there is no cable connection between the transmitter and the receiver, some modifications to the setup are required to ensure precise frequency and time synchronization.

A. Frequency synchronization

As a possible solution for the frequency synchronization, we employ a Global Positioning System (GPS) unit with a Rubidium frequency standard at the transmitter and the receiver [24]. Locked to geostationary satellites, both GPS units output one pulse per second (PPS). In the next step, the long-term-stable PPS pulses are short-term-stabilized by the Rubidium frequency standard. The Rubidium frequency standard ensures a maximum frequency deviation of $\pm 5 \cdot 10^{-11}$, which leads to a worst case deviation between transmitter and receiver of $\Delta f = 10 \cdot 10^{-11} \times 5.9 \text{ GHz} = 0.59 \text{ Hz}$. This frequency deviation is negligible when compared to the employed subcarrier spacing of the OFDM signal (see Tab. I).

At the transmitter, the 10MHz output of the Rubidium standard is directly provided as a reference input of the arbitrary waveform generator.

At the receiver side, the 10MHz reference signal is out of range of the employed signal analyzer. Therefore, the 10MHz reference output of the rubidium is fed to the 200MHz oscillator. Finally, the 200MHz output of the oscillator is given as a reference input of the signal analyzer. Thereby, we ensure a precise frequency synchronization between the transmitter and the receiver.

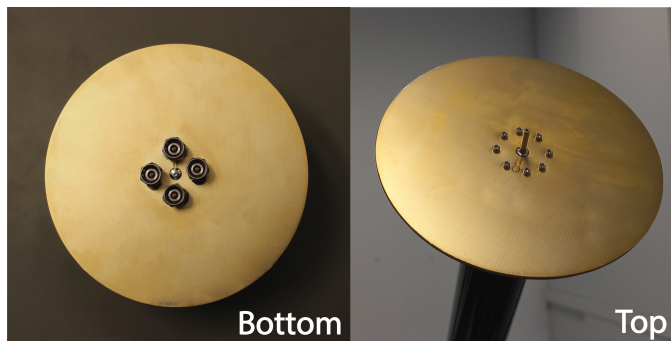


Fig. 2. 5.9GHz rotating transmit antenna: connectors on the bottom side (left) and monopole antenna on the top side (right).



Fig. 3. 5.9GHz static receive antenna placed on the roof of a building in the center of Vienna.

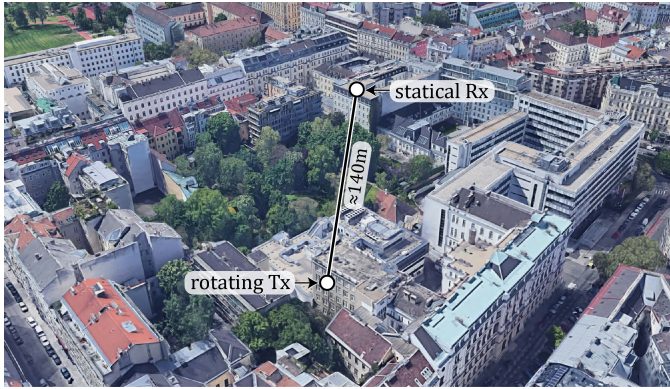


Fig. 4. Measured urban environment. The transmit antenna is located inside the building in a rich scattering environment. The receive antenna is static on the roof of another building.

B. Time synchronization

There is a similar issue with time synchronization. The rotary unit is equipped with a self-built trigger unit, which can repeatably trigger a measurement at an adjustable and precisely defined arm angle, once per revolution. A trigger signal is fed to the arbitrary waveform generator at the transmitter through a cable. Unfortunately, there is no cable connection between the transmitter's trigger unit and the receiver's signal analyzer to provide precise time synchronization.

We solve the time-synchronization problem in the following way. Firstly, we record $N = (t_r + t_s) f_s$ samples at the receiver, where t_r represents the duration of a single rotation, t_s is the duration of the transmit signal and f_s is the sampling rate at the receiver. In this way, we ensure at least one whole transmit signal to be recorded by the receiver. After that, we exploit good auto-correlation properties of the Zadoff-Chu sequence in the time domain [25] to find the beginning of the received channel-sounding signal within the recorded samples. More precisely, detecting the start of the signal can be accomplished by calculating the correlation of the Zadoff-Chu sequence with the received signal. Thereby, we ensure precise time synchronization between the transmitter and the receiver.

C. Measurement Methodology

We conduct measurements at low speeds (25 km/h and 50 km/h) and high speeds (100 km/h and 125 km/h) at the 5.9 GHz center frequency with channel sounding parameters provided in Tab. I. As a channel-sounding sequence, we transmit 50 000 identical Zadoff-Chu OFDM symbols [23]. The transmission is initiated using the self-built trigger unit at a precisely defined arm angle. More precisely, when the rotating arm reaches an angular position of -40° , the trigger signal is fed to the arbitrary waveform generator and the transmission starts. In order to achieve the comparability of measurement results, we adjust an OFDM symbol duration to the transmitter's velocities. Since the symbol duration is inversely proportional to the subcarrier spacing, the adjustment

TABLE I
CHANNEL SOUNDING PARAMETERS

Parameter	Scenario			
	A	B	C	D
Transmitter Velocity (km/h)	25	50	100	125
Number of Subcarriers	400	200	100	80
Subcarrier Spacing (kHz)	250	500	1000	1250
Symbol Duration (μ s)	4	2	1	0.8
Snapshot Duration (μ s)	400	200	100	80
Measurement Duration (ms)	200	100	50	40
Maximum Doppler Shift (Hz)	136	273	546	683
Carrier Frequency (GHz)	5.9			
Bandwidth (MHz)	100			
Delay Resolution (ns)	10			
Number of Snapshots	500			
Symbols per Snapshot	100			

is achieved by choosing different subcarrier spacings for different transmitter's velocities. We choose a low subcarrier spacing of 250 kHz and 500 kHz for low transmitter's velocities of 25 km/h and 50 km/h, respectively. On the other hand, for high-velocity scenarios of 100 km/h and 125 km/h, we choose a high subcarrier spacing of 1 MHz and 1.25 MHz, respectively. Thus, the rotating transmit antenna moves along the same trace (arc segment) from -40° to 40° for each measured scenario, while the receive antenna maintains the same position. Thereby, we achieve the comparability of the measurement results in terms of fading environment and channel statistics.

Similar as in [26], we assume the wireless channel is constant in time for the duration of 100 symbols. Therefore, we group each received sequence into 500 snapshots of 100 symbols each. To counteract the multi-path propagation effects of the wireless channel, we use the first OFDM symbol of each snapshot as a cyclic prefix and discard it. After that, we perform averaging of the remaining 99 symbols of each snapshot to improve the signal-to-noise ratio (SNR) followed by a least-squares estimation of the wireless channel for all subcarriers. As a result, we obtain a time-variant channel transfer function for discrete-time (snapshots) and frequency (subcarriers).

IV. RESULTS

Using the concept described in [27], we estimate the local scattering function of the wireless channel for each scenario. The estimation of the local scattering function is performed under the assumption that the wireless channel is locally stationary within a window of approximately six wavelengths of motion (100 snapshots for the 5.9 GHz case) and over the entire frequency range. We calculate the local scattering function for each window consisting of 100 snapshots and all subcarriers. After that, we average the local scattering function of each window to produce the average local scattering function as a result.

In order to simplify the comparison of the results, we perform two normalizations of the average local scattering function. Firstly, we adjust the Doppler shift such that the average local scattering function does not depend on differ-

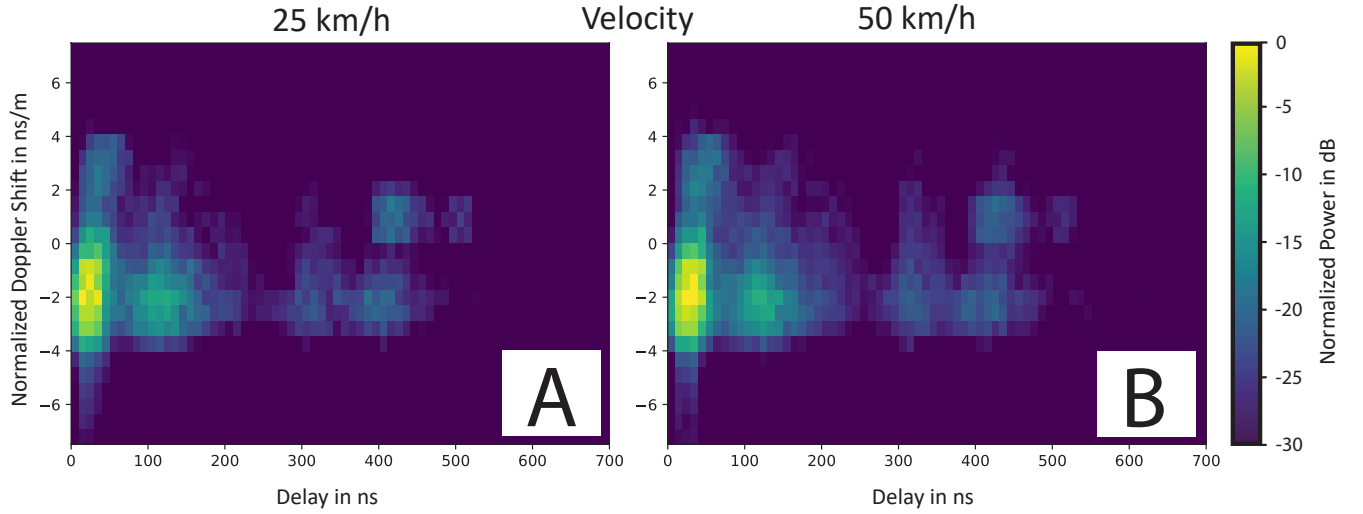


Fig. 5. Low-velocity scenarios: Measurement results in terms of the normalized average local scattering function for velocities of 25 km/h and 50 km/h.

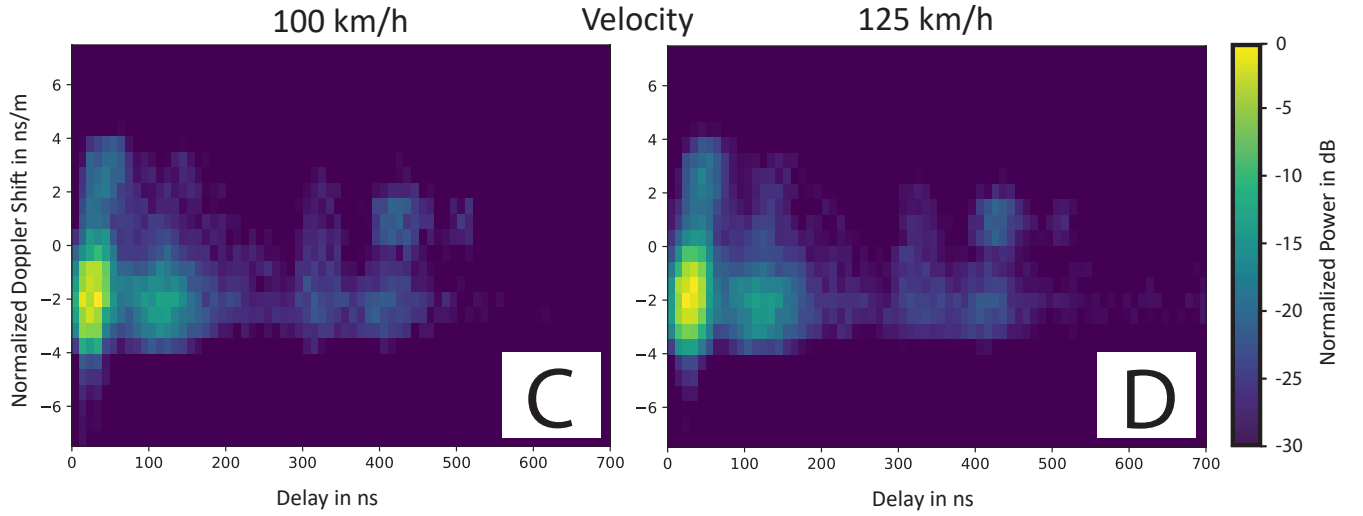


Fig. 6. High-velocity scenarios: Measurement results in terms of the normalized average local scattering function for velocities of 100 km/h and 125 km/h.

ent transmitter's velocities. The adjustment is performed by introducing the normalized Doppler shift ν_N

$$\nu_N = \frac{\nu}{f_c v_{Tx}}, \quad (2)$$

where f_c denotes the center frequency, v_{Tx} represents the transmitter's velocity and ν is the Doppler shift. Secondly, we normalize the intensity of the average local scattering function such that 0 dB is the greatest occurring value in each scenario. Furthermore, we limit the displayed delay to 700 ns, since no significant components were observed for larger delays. The results for low-velocity and high-velocity cases are shown in Fig. 5 and Fig. 6, respectively.

We can observe that the normalized average local scattering function does not depend on the transmitter's velocity. More precisely, all scenarios look alike, although they are measured at different velocities. Furthermore, we can observe that

the scattering environment is frequency selective. Since the transmitter moves away from the receiver, the main Doppler component is negative.

V. CONCLUSION

In this paper, we adapted the measurement testbed for performing measurements at the 5.9 GHz band. We are able to conduct channel-sounding measurements in an indoor-to-outdoor high-mobility environment. We showed results in terms of the normalized average local scattering function. The normalized average local scattering functions have strong similarities, although the measurements are performed at different velocities. The results indicate that the proposed methodology to conduct repeatable measurements at different velocities in a controlled environment works correctly.

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